Eunsue Choi

Updated Sep, 2026

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Research interests

Computational Imaging, Wave Optics, Metasurfaces, Holographic Displays, End-to-end optimization, Computer Vision, Computer Graphics, Cameras, Displays, lenses and Sensing, VR/AR

Education

POSTECH Sep.2022 – Present
MS-PhD in Computer Science and Engineering Pohang, S.Korea
Advisor: Dr. Seung-Hwan Baek

POSTECH Mar.2018 – Aug.2022
BS in Computer Science and Engineering Pohang, S.Korea
Graduated with Magna Cum Laude

Experience

Research Scientist Intern Jun.2026 – Oct.2026
Display Systems Research Team Redmond, WA
Meta Reality Labs

Research Scientist Intern Jun.2025 – Nov.2025
Display Systems Research Team Redmond, WA
Meta Reality Labs

Publications

[1] Learned π Metalens for Wide-Field-of-View Near-Infrared Imaging
Eunsue Choi*, Gyeongtae Kim*, Eunji Lee*, Seungwoo Yoon, Youngmo Jeong, Woohyeok Kim, Minseok Choi, Sunghyun Cho, Junsuk Rho+, Seung-Hwan Baek+
ACM Transactions on Graphics (TOG) (in press), 2026

[2] De-occluding broadband metalens [Link]
Seungwoo Yoon, Dohyun Kang, **Eunsue Choi**, Sohyun Lee, Seoyeon Kim, Minho Choi, Hyeonsu Heo, Dong-Ha Shin, Suha Kwak, Arka Majumdar, Junsuk Rho, Seung-Hwan Baek
Nature Communications, 2026

[3] Limitations of data-driven spectral reconstruction: an optics-aware analysis [Link]
Qiang Fu, Matheus Souza, **Eunsue Choi**, Suhyun Shin, Seung-Hwan Baek, Wolfgang Heidrich
IEEE Transactions on Computational Imaging, 2025

[4] 360° Structured Light with Learned Metasurfaces [Link]
Eunsue Choi*, Gyeongtae Kim*, JooyeongYun, Yujin Jeon, Junsuk Rho+, Seung-Hwan Baek+
Nature Photonics, 2024.
* Featured in Nature Computational Science's special issue, highlighted in editorial and research highlights

[5] Spectral and Polarization Vision: Spectro-polarimetric Real-world Dataset [Link]
Yujin Jeon*, **Eunsue Choi***, Youngchan Kim, Yunseong Moon, Khalid Omer, Felix Heide, Seung-Hwan Baek, (* equal contributions)
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
* Highlight paper (top 11.9%)

Patents

Electronic Device for Image Processing, Operating Method Thereof, and Storage Medium
Youngmo Jeong, Seung-Hwan Baek, **Eunsue Choi**, et al.
Korean Patent Publication No. 10-2026-0092928 (2026)

Display Device Projecting Light 360 Degrees Through Trained Metasurface and Performing Three-Dimensional Imaging, and Controlling Method Thereof
Eunsue Choi, et al.
Korean Patent Publication No. 10-2025-0125853 (2025)
U.S. Patent Application No. 18/776,856 (Pending, 2024)

Honors and Awards

NRF PhD Fellowship, National Research Foundation of Korea, 2025
Postechian Fellowship, POSTECH, 2025
Outstanding M.S. Thesis Award, Korea Computer Graphics Society 2024
Alchemist R&D Fellowship, POSTECH 2024
Outstanding Interdisplinary Research Award, POSTECH 2024

Talks

Seeing beyond conventional light with wave optics
Korea Computer Graphics Society, Award Talk July.2024

Teaching experience

Teaching assistant, Data Structure, POSTECH Spring 2025
Teaching assistant, Data Structure, POSTECH Spring 2024
Teaching assistant, Data Structure, POSTECH Fall 2023